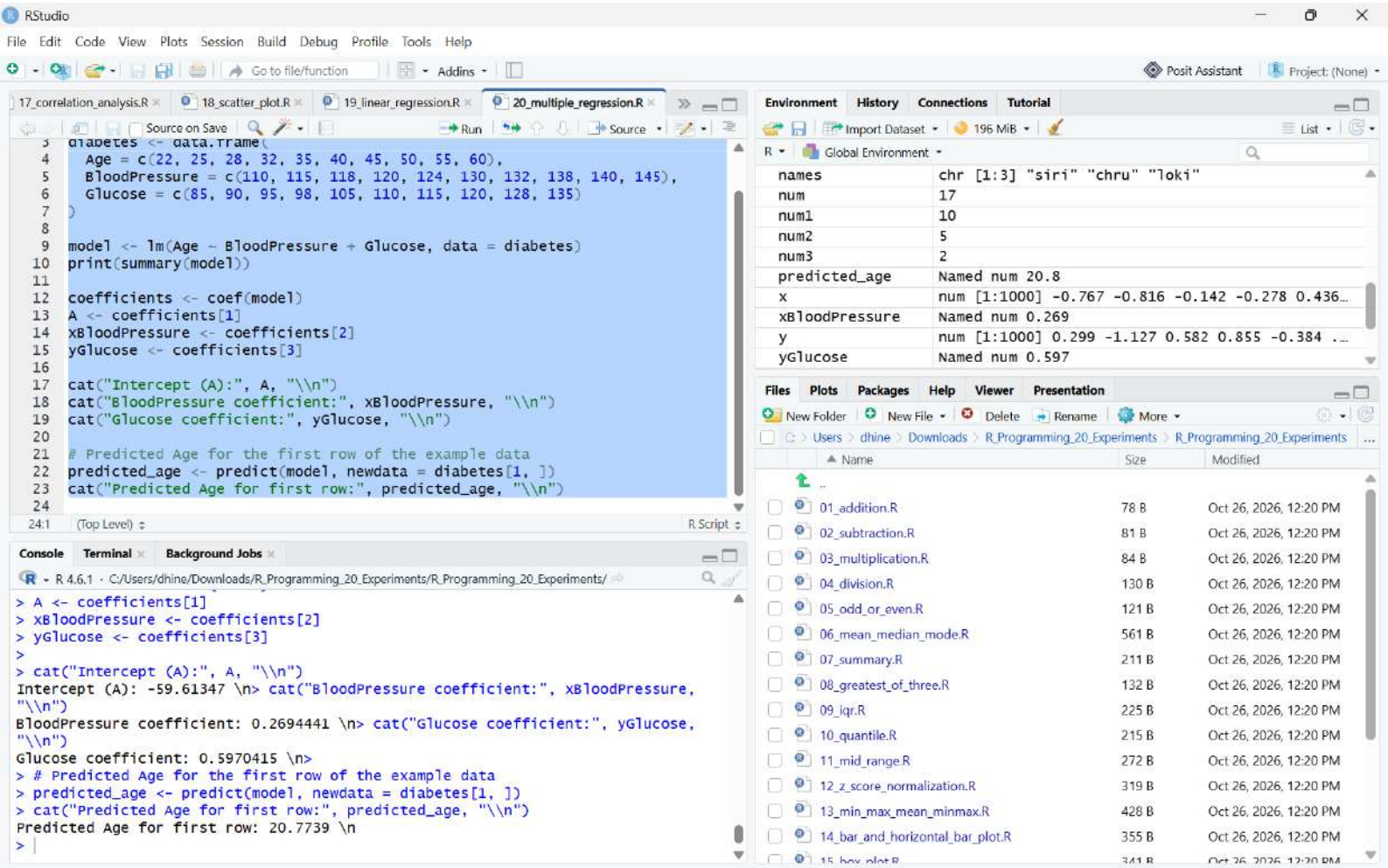

RStudio
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Go to file/function
Addins
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Project: (None)
17_correlation_analysis.R
18_scatter_plot.R
19_linear_regression.R
20_multiple_regression.R
Source on Save
Run
Source
diabetes <- data.frame(
Age = c(22, 25, 28, 32, 35, 40, 45, 50, 55, 60),
BloodPressure = c(110, 115, 118, 120, 124, 130, 132, 138, 140, 145),
Glucose = c(85, 90, 95, 98, 105, 110, 115, 120, 128, 135)
)
model <- lm(Age ~ BloodPressure + Glucose, data = diabetes)
print(summary(model))
coefficients <- coef(model)
A <- coefficients[1]
xBloodPressure <- coefficients[2]
yGlucose <- coefficients[3]
cat("Intercept (A):", A, "\\n")
cat("BloodPressure coefficient:", xBloodPressure, "\\n")
cat("Glucose coefficient:", yGlucose, "\\n")
Predicted Age for the first row of the example data
predicted_age <- predict(model, newdata = diabetes[1,])
cat("Predicted Age for first row:", predicted_age, "\\n")
24:1
(Top Level)
R Script
Console
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Background Jobs
R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/
> A <- coefficients[1]
> xBloodPressure <- coefficients[2]
> yGlucose <- coefficients[3]
> cat("Intercept (A):", A, "\\n")
Intercept (A): -59.61347 \n> cat("BloodPressure coefficient:", xBloodPressure, "\\n")
BloodPressure coefficient: 0.2694441 \n> cat("Glucose coefficient:", yGlucose, "\\n")
Glucose coefficient: 0.5970415 \n>
> # Predicted Age for the first row of the example data
> predicted_age <- predict(model, newdata = diabetes[1,])
> cat("Predicted Age for first row:", predicted_age, "\\n")
Predicted Age for first row: 20.7739 \n
Environment
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Import Dataset
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R
Global Environment
names chr [1:3] "siri" "chru" "loki"
num 17
num1 10
num2 5
num3 2
predicted_age Named num 20.8
x num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436…
xBloodPressure Named num 0.269
y num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 .…
yGlucose Named num 0.597
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Name
Size
Modified
01_addition.R 78 B Oct 26, 2026, 12:20 PM
02_subtraction.R 81 B Oct 26, 2026, 12:20 PM
03_multiplication.R 84 B Oct 26, 2026, 12:20 PM
04_division.R 130 B Oct 26, 2026, 12:20 PM
05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 561 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_iqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 428 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 355 B Oct 26, 2026, 12:20 PM

RStudio

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16_histogram.R | 17_correlation_analysis.R | 18_scatter_plot.R | 19_linear_regression.R

```
# Experiment 19: Linear Regression
# Example data is used instead of an external diabetes dataset.
diabetes <- data.frame(
  Age = c(22, 25, 28, 32, 35, 40, 45, 50, 55, 60),
  BloodPressure = c(110, 115, 118, 120, 124, 130, 132, 138, 140, 145)
)

model <- lm(BloodPressure ~ Age, data = diabetes)
print(summary(model))

plot(diabetes$Age, diabetes$BloodPressure,
     col = "darkgreen", pch = 19,
     main = "Linear Regression Analysis",
     xlab = "Age", ylab = "Blood Pressure")
abline(model, col = "red", lwd = 2)

```

16:1 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/

```
Age          0.88922    0.03298   26.96 3.86e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.288 on 8 degrees of freedom
Multiple R-squared:  0.9891,    Adjusted R-squared:  0.9878
F-statistic: 726.8 on 1 and 8 DF,  p-value: 3.859e-09

>
> plot(diabetes$Age, diabetes$BloodPressure,
+      col = "darkgreen", pch = 19,
+      main = "Linear Regression Analysis",
+      xlab = "Age", ylab = "Blood Pressure")
> abline(model, col = "red", lwd = 2)
>
```

Environment | History | Connections | Tutorial

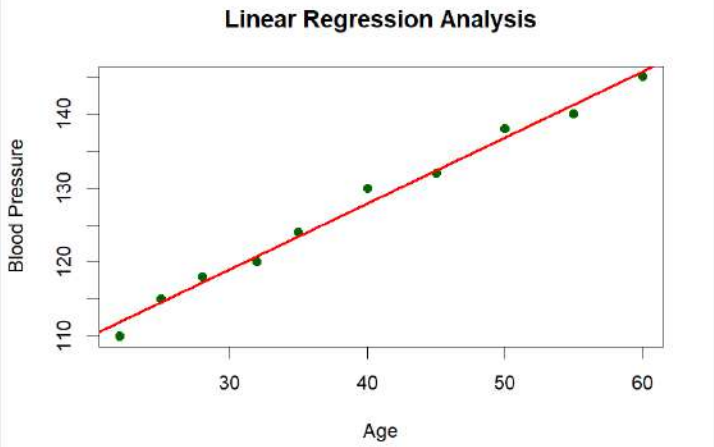
Import Dataset · 194 MiB · List

R · Global Environment

model	List of 12
Values	
a	num [1:5] 55 67 89 80 90
age	num [1:3] 23 24 25
age_group	chr [1:12] "Young" "Young" "Middle" "Middle" "…
age_mean	30.2
age_sd	7.59605160593318
frequency_table	'table' int [1:3, 1:3] 2 2 0 0 1 3 2 1 1
greatest	42
insulin_group	chr [1:12] "Low" "Low" "Normal" "High" "High" …

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ar_plot.R | 15_box_plot.R | 16_histogram.R | 17_correlation_analysis.R | 18_scatter_plot.R

Source on Save | Run | Source

```
1  # Experiment 18: Scatter Plot
2  set.seed(9)
3  x <- rnorm(1000)
4  y <- rnorm(1000)
5
6  par(mfrow = c(1, 2))
7  smoothScatter(y - x, main = "Smoothed Density of y - x",
8                xlab = "Index", ylab = "y - x")
9  smoothScatter(x, y, main = "Scatter Plot", xlab = "x", ylab = "y")
10 par(mfrow = c(1, 1))
11
```

11:1 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/

```
Warning message:
In chisq.test(frequency_table) : Chi-squared approximation may be incorrect

> # Experiment 18: Scatter Plot
> set.seed(9)
> x <- rnorm(1000)
> y <- rnorm(1000)
>
> par(mfrow = c(1, 2))
> smoothScatter(y - x, main = "Smoothed Density of y - x",
+               xlab = "Index", ylab = "y - x")
> smoothScatter(x, y, main = "Scatter Plot", xlab = "x", ylab = "y")
> par(mfrow = c(1, 1))
>
```

Environment | History | Connections | Tutorial

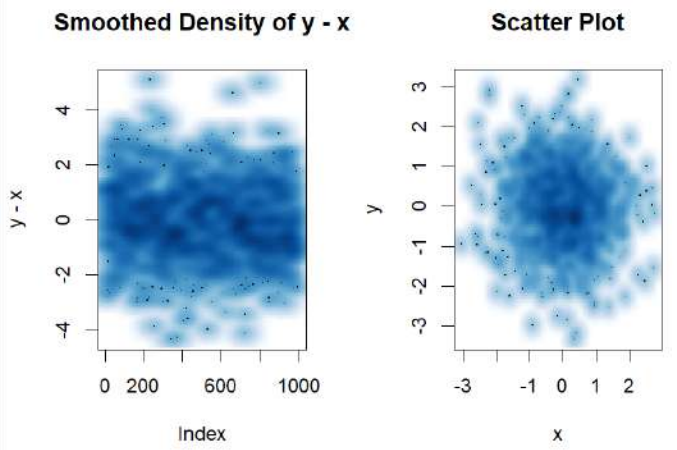
Import Dataset | 209 MiB | List

R | Global Environment

mid_range	24
min_max	num [1:5] 0 0.211 0.474 0.737 1
minimum	21
names	chr [1:3] "siri" "chru" "loki"
num	17
num1	10
num2	5
num3	2
x	num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436…
y	num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 …

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Go to file/function | Addins | Posit Assistant | Project: (None)

14_bar_and_horizontal_bar_plot.R | 15_box_plot.R | 16_histogram.R | 17_correlation_analysis.R

Source on Save | Run | Source

```
1  # Experiment 17: Association Analysis using a Chi-square Test
2  # Example categorical data is used instead of an external Excel file.
3  age_group <- c("Young", "Young", "Middle", "Middle", "Older",
4                 "Older", "Young", "Middle", "Older", "Young",
5                 "Middle", "Older")
6  insulin_group <- c("Low", "Low", "Normal", "High", "High",
7                     "Normal", "Normal", "High", "Low", "Low",
8                     "Normal", "High")
9
10 frequency_table <- table(age_group, insulin_group)
11 print(frequency_table)
12 print(chisq.test(frequency_table))
13
```

13:1 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/

```
+                 "Normal", "High")
>
> frequency_table <- table(age_group, insulin_group)
> print(frequency_table)
           insulin_group
age_group High Low Normal
   Middle    2   0      2
   Older     2   1      1
   Young     0   3      1
> print(chisq.test(frequency_table))

        Pearson's Chi-squared test

data:  frequency_table
X-squared = 6, df = 4, p-value = 0.1991
```

Environment | History | Connections | Tutorial

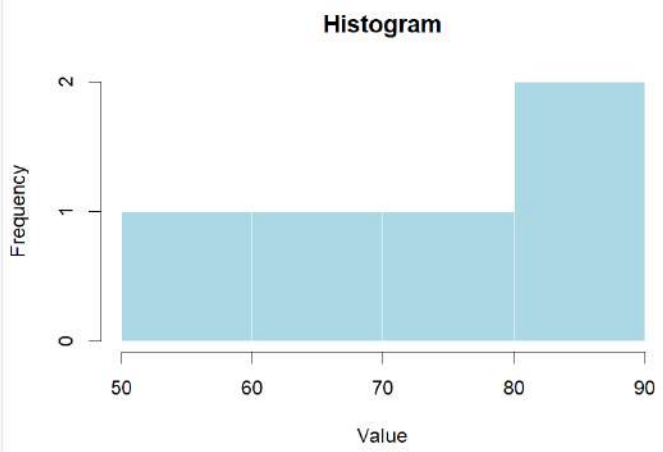
Import Dataset | 188 MiB | List

R | Global Environment

age_group	chr [1:12] "Young" "Young" "Middle" "Middle" "…
age_mean	30.2
age_sd	7.59605160593318
frequency_table	'table' int [1:3, 1:3] 2 2 0 0 1 3 2 1 1
greatest	42
insulin_group	chr [1:12] "Low" "Low" "Normal" "High" "High" …
marks	num [1:3] 88 78 25
maximum	40
mean_age	30.2
mid_range	24

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Go to file/function | Addins | Posit Assistant | Project: (None)

_min_max_mean_minmax.R | 14_bar_and_horizontal_bar_plot.R | 15_box_plot.R | 16_histogram.R

Source on Save | Run | Source

```
1  # Experiment 16: Histogram
2  a <- c(55, 67, 89, 80, 90)
3  hist(a, main = "Histogram", xlab = "Value", col = "lightblue",
4       border = "white")
5
```

5:1 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.4.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/

```
> # Experiment 15: Histogram and Box Plot
> names <- c("siri", "chru", "loki")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
>
> par(mfrow = c(1, 2))
> hist(df$age, main = "Age Histogram", xlab = "Age", col = "lightblue")
> boxplot(df$age, main = "Age Box Plot", ylab = "Age", col = "lightgreen")
> par(mfrow = c(1, 1))
> # Experiment 16: Histogram
> a <- c(55, 67, 89, 80, 90)
> hist(a, main = "Histogram", xlab = "Value", col = "lightblue",
+      border = "white")
>
```

Environment | History | Connections | Tutorial

Import Dataset | 188 MiB | List

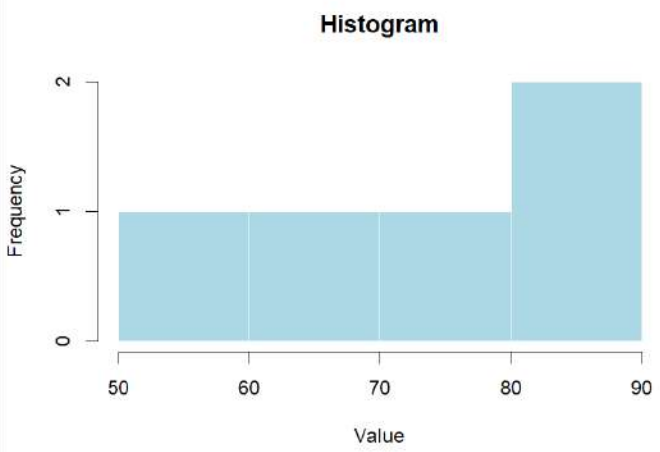
R | Global Environment

Values

a	num [1:5] 55 67 89 80 90
age	num [1:3] 23 24 25
age_mean	30.2
age_sd	7.59605160593318
greatest	42
marks	num [1:3] 88 78 25
maximum	40
mean_age	30.2
mid_range	24

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rmalization.R | 13_min_max_mean_minmax.R | 14_bar_and_horizontal_bar_plot.R | 15_box_plot.R

Source on Save | Run | Source

```
1  # Experiment 15: Histogram and Box Plot
2  names <- c("siri", "chru", "loki")
3  age <- c(23, 24, 25)
4  marks <- c(88, 78, 25)
5  df <- data.frame(names, age, marks)
6
7  par(mfrow = c(1, 2))
8  hist(df$age, main = "Age Histogram", xlab = "Age", col = "lightblue")
9  boxplot(df$age, main = "Age Box Plot", ylab = "Age", col = "lightgreen")
10 par(mfrow = c(1, 1))
11
```

11:1 (Top Level) | R Script

Console | Terminal | Background Jobs

R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/

```
+         names.arg = paste0("Item ", seq_along(a)))
> barplot(a, horiz = TRUE, main = "Horizontal Bar Plot",
+         xlab = "Value", names.arg = paste0("Item ", seq_along(a)))
> par(mfrow = c(1, 1))
> # Experiment 15: Histogram and Box Plot
> names <- c("siri", "chru", "loki")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
>
> par(mfrow = c(1, 2))
> hist(df$age, main = "Age Histogram", xlab = "Age", col = "lightblue")
> boxplot(df$age, main = "Age Box Plot", ylab = "Age", col = "lightgreen")
> par(mfrow = c(1, 1))
>
```

Environment | History | Connections | Tutorial

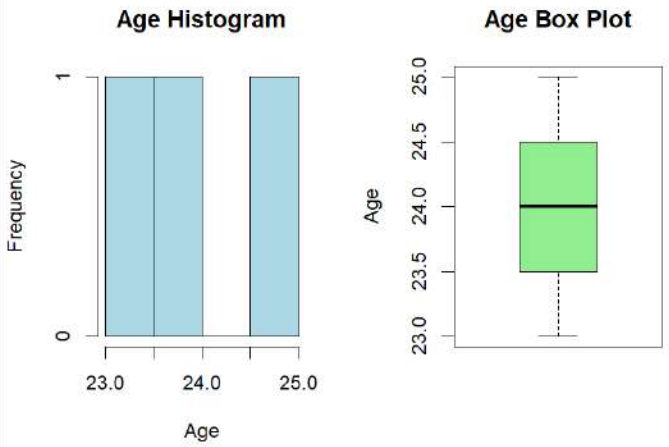
Import Dataset | 188 MiB | List

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age_mean	30.2
age_sd	7.59605160593318
greatest	42
marks	num [1:3] 88 78 25
maximum	40
mean_age	30.2
mid_range	24
min_max	num [1:5] 0 0.211 0.474 0.737 1
minimum	21
names	chr [1:3] "siri" "chru" "loki"

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12_z_score_normalization.R | 13_min_max_mean_minmax.R | 14_bar_and_horizontal_bar_plot.R

Source on Save | Run | Source

```
1  # Experiment 14: Vertical and Horizontal Bar Plots
2  a <- c(55, 67, 89, 80, 90)
3
4  par(mfrow = c(1, 2))
5  barplot(a, main = "Bar Plot", xlab = "Item", ylab = "Value",
6          names.arg = paste0("Item ", seq_along(a)))
7  barplot(a, horiz = TRUE, main = "Horizontal Bar Plot",
8          xlab = "Value", names.arg = paste0("Item ", seq_along(a)))
9  par(mfrow = c(1, 1))
10
```

10:1 (Top Level) R Script

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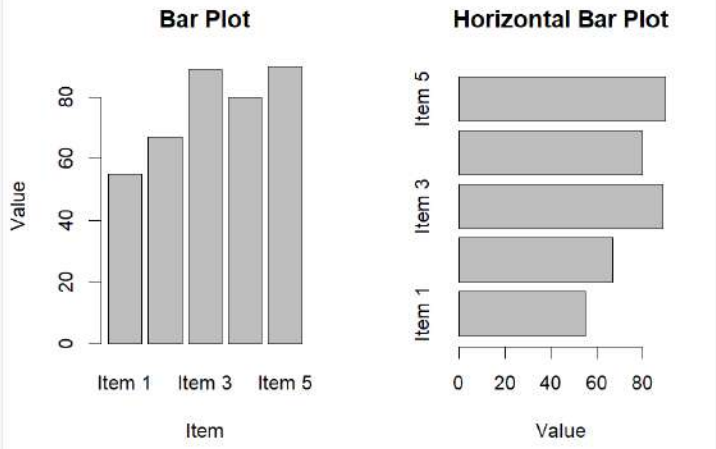
R | Global Environment

Values

a	num [1:5] 55 67 89 80 90
age	num [1:5] 21 25 30 35 40
age_mean	30.2
age_sd	7.59605160593318
greatest	42
marks	num [1:3] 88 78 25
maximum	40
mean_age	30.2
mid_range	24

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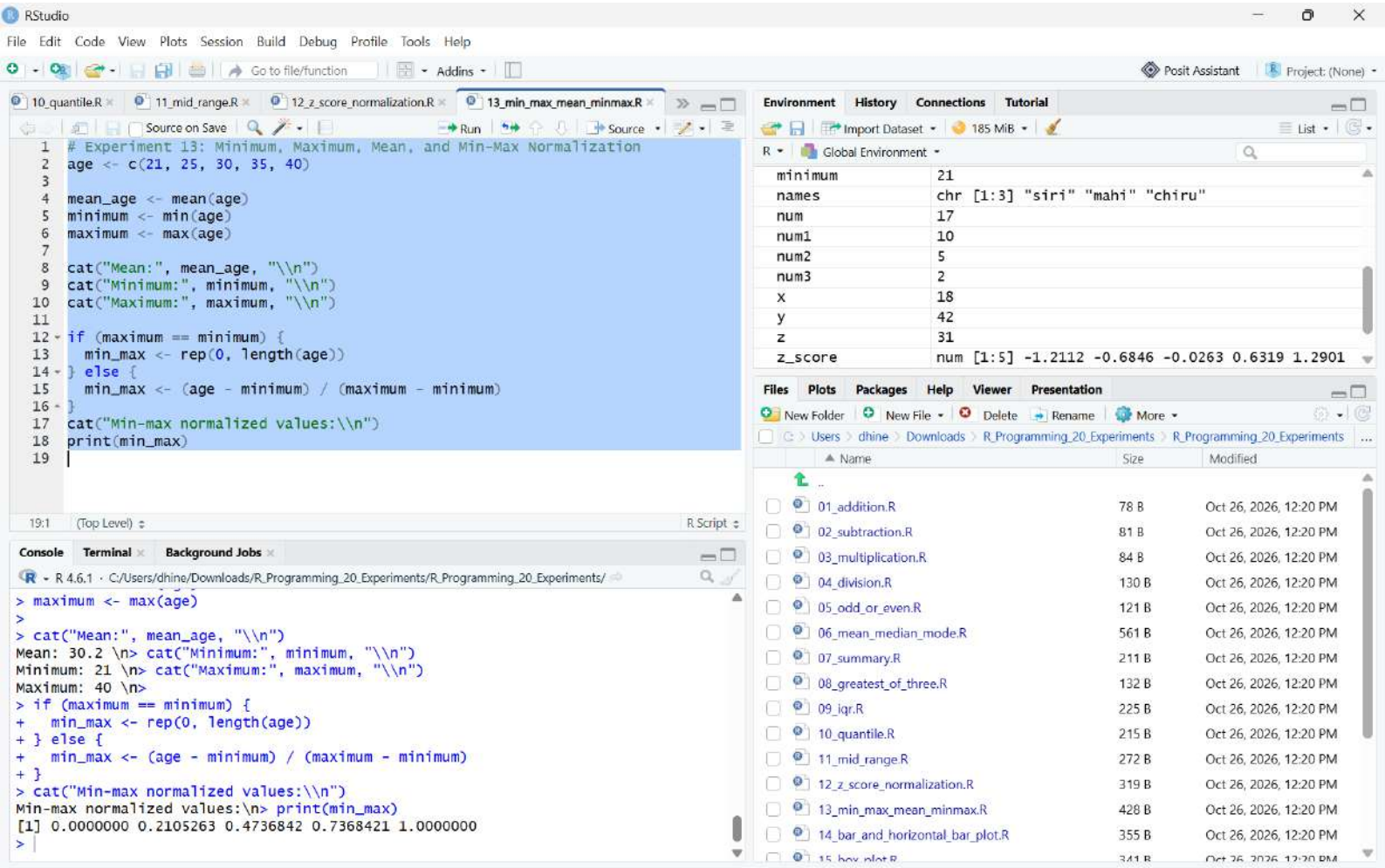
Zoom | Fit to Pane | Export | Publish

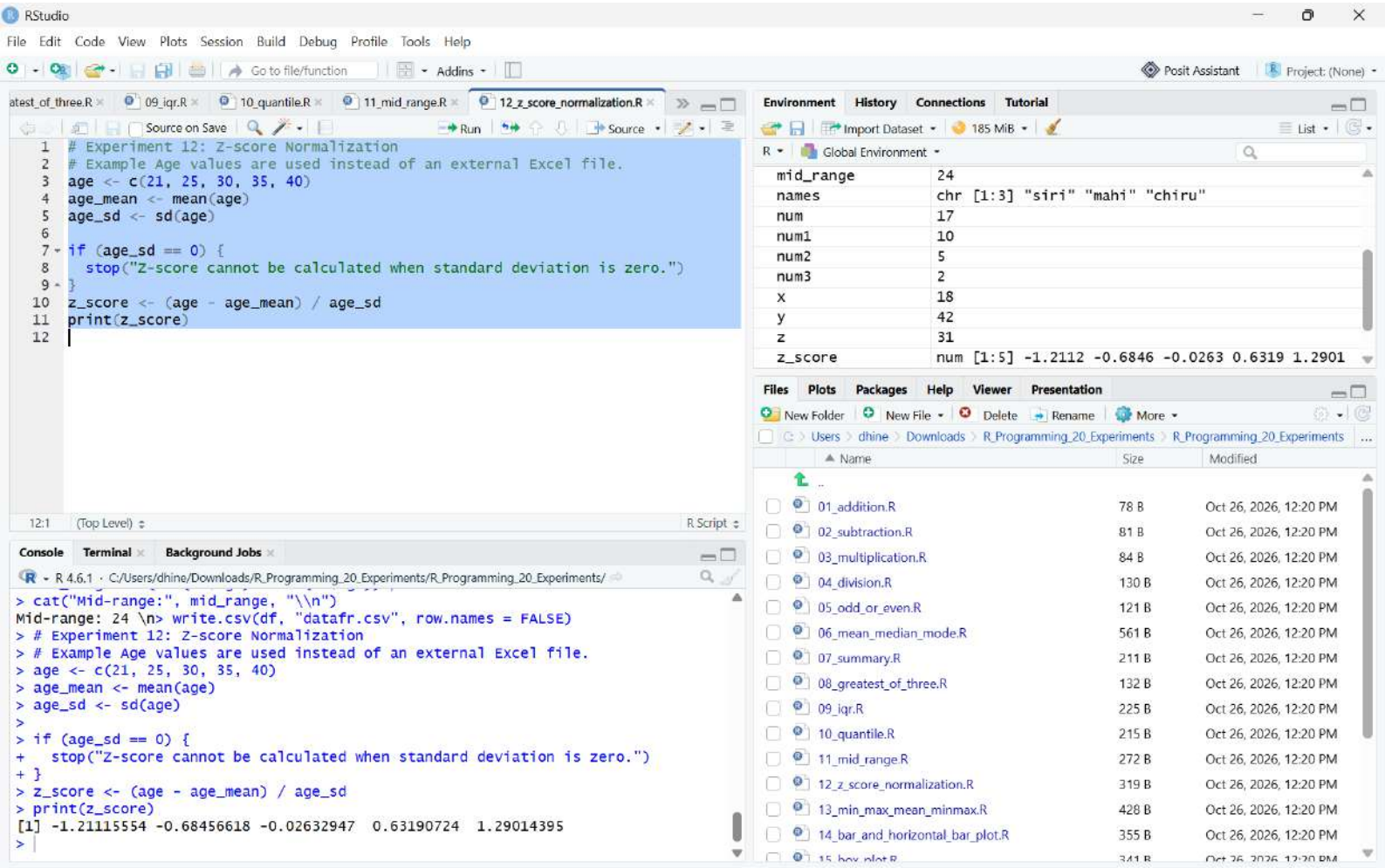


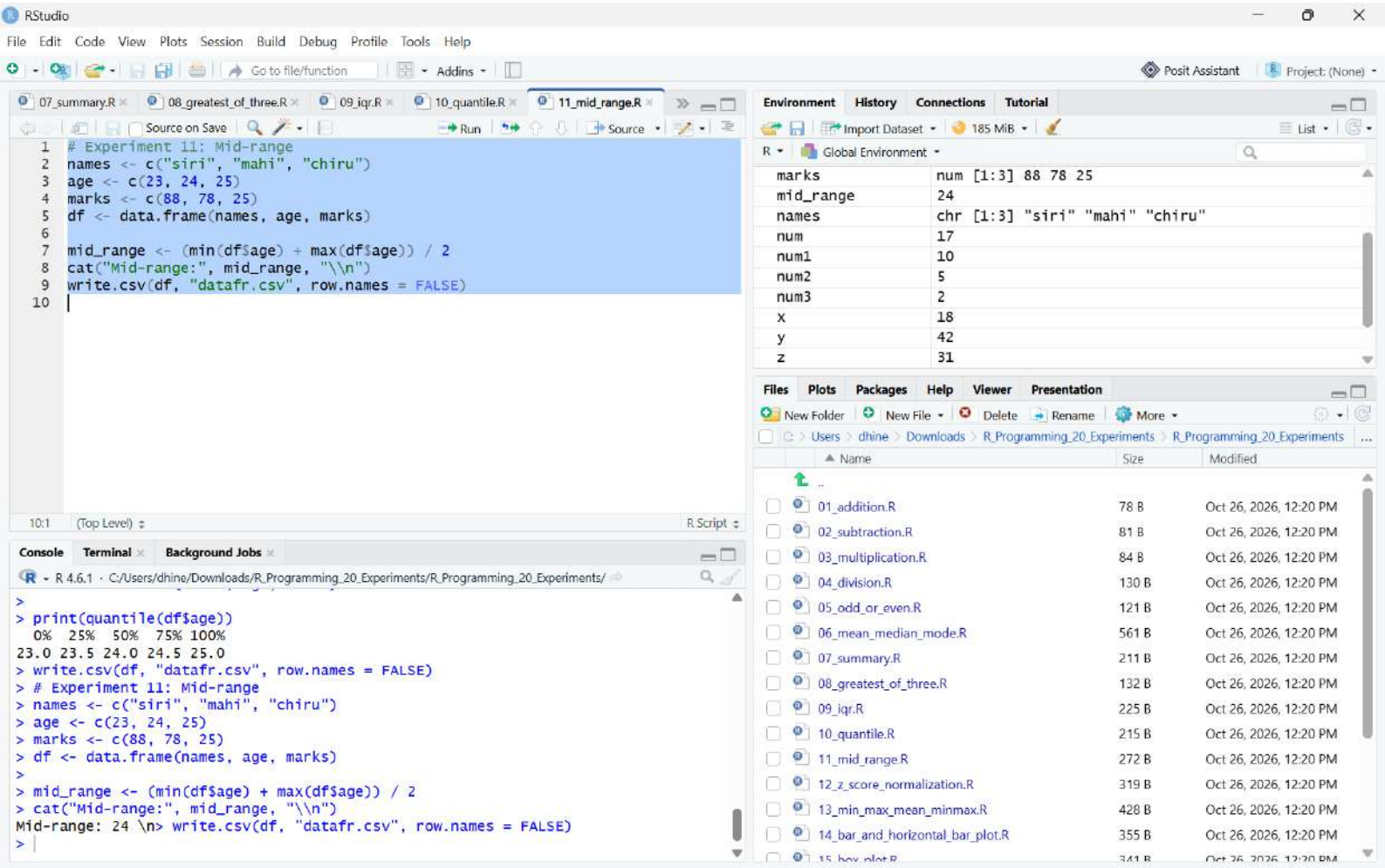
Console | Terminal | Background Jobs

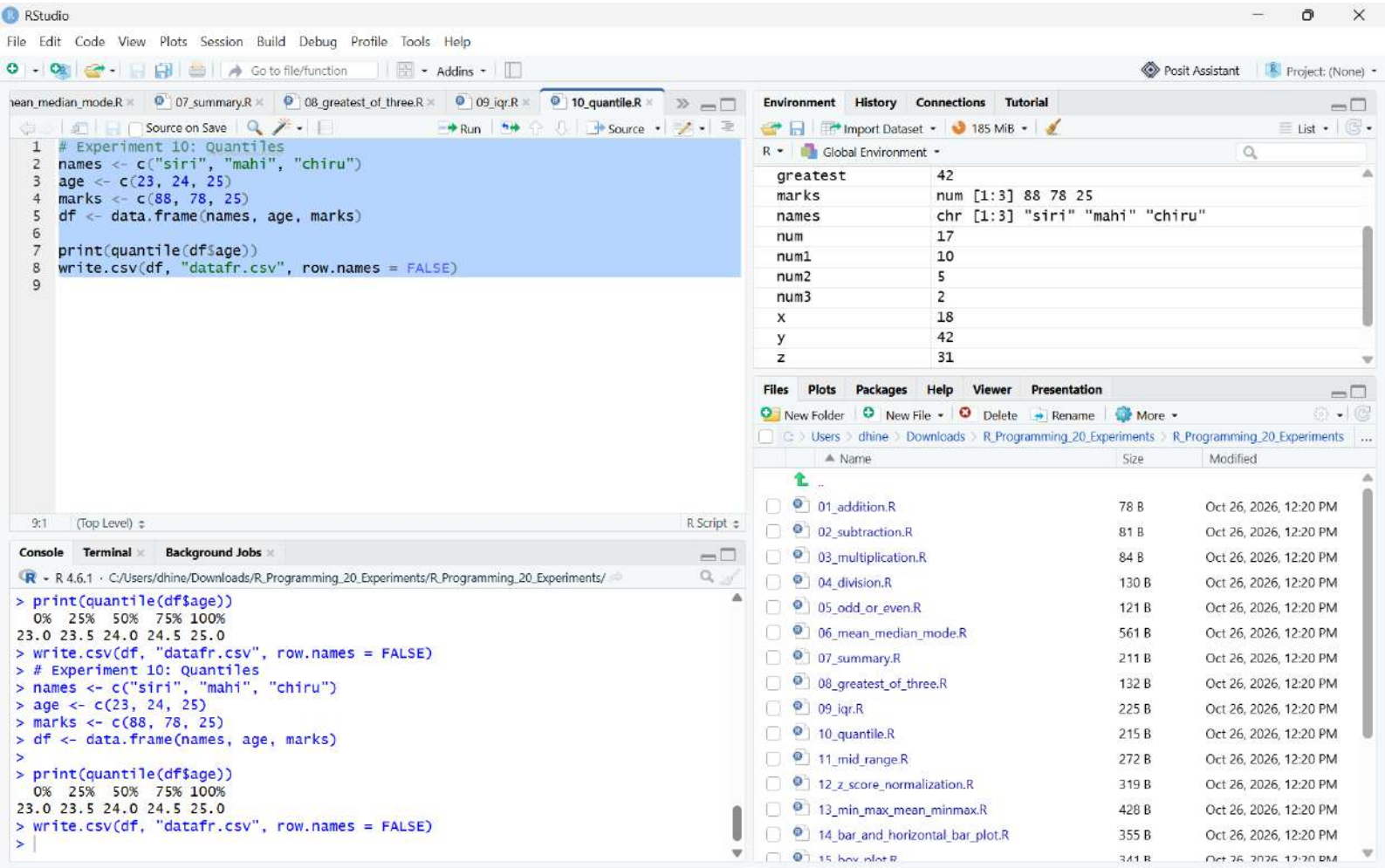
R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/

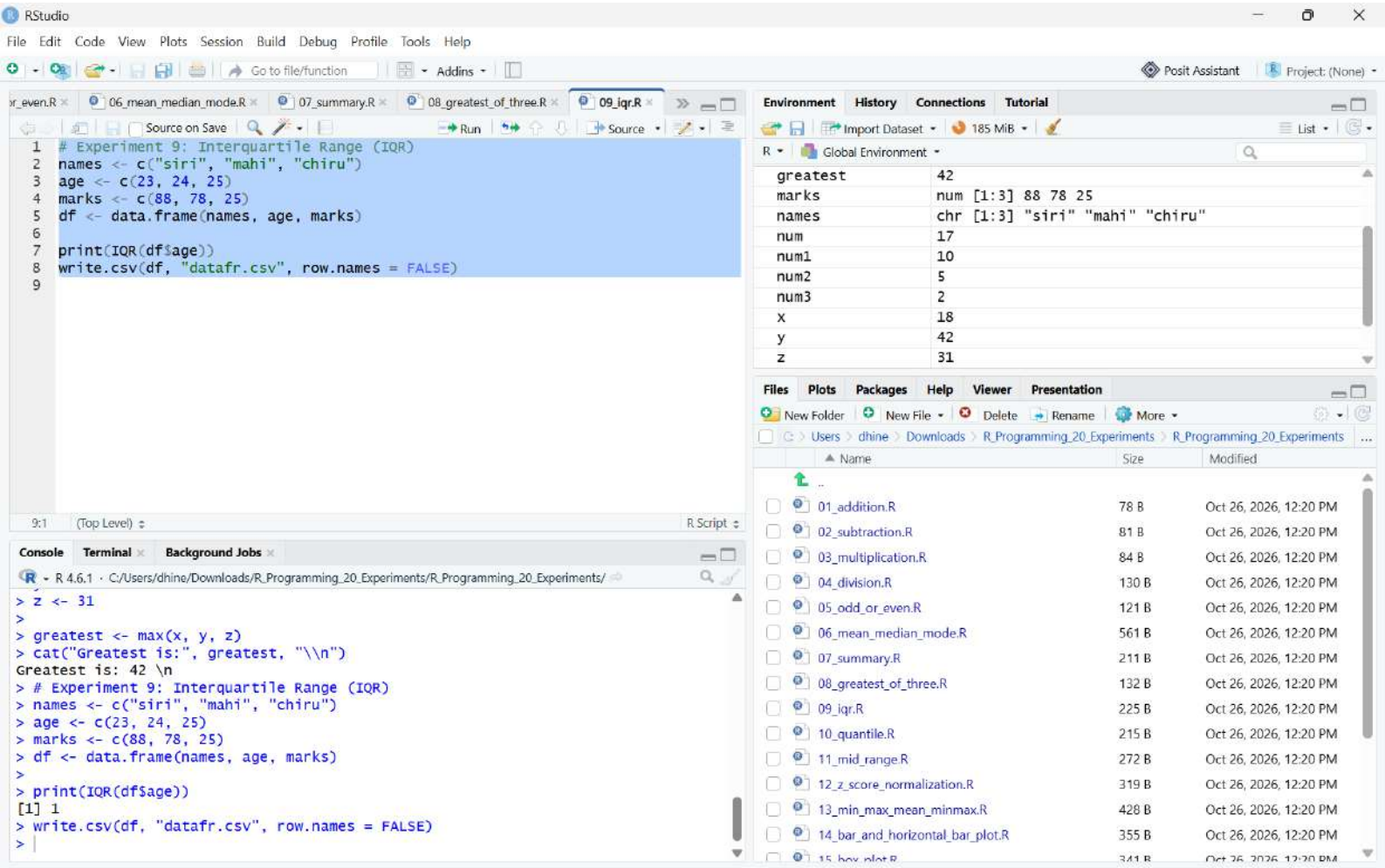
```
+   min_max <- (age - minimum) / (maximum - minimum)
+ }
> cat("Min-max normalized values:\\n")
Min-max normalized values:\n> print(min_max)
[1] 0.0000000 0.2105263 0.4736842 0.7368421 1.0000000
> # Experiment 14: Vertical and Horizontal Bar Plots
> a <- c(55, 67, 89, 80, 90)
>
> par(mfrow = c(1, 2))
> barplot(a, main = "Bar Plot", xlab = "Item", ylab = "Value",
+         names.arg = paste0("Item ", seq_along(a)))
> barplot(a, horiz = TRUE, main = "Horizontal Bar Plot",
+         xlab = "Value", names.arg = paste0("Item ", seq_along(a)))
> par(mfrow = c(1, 1))
>
```

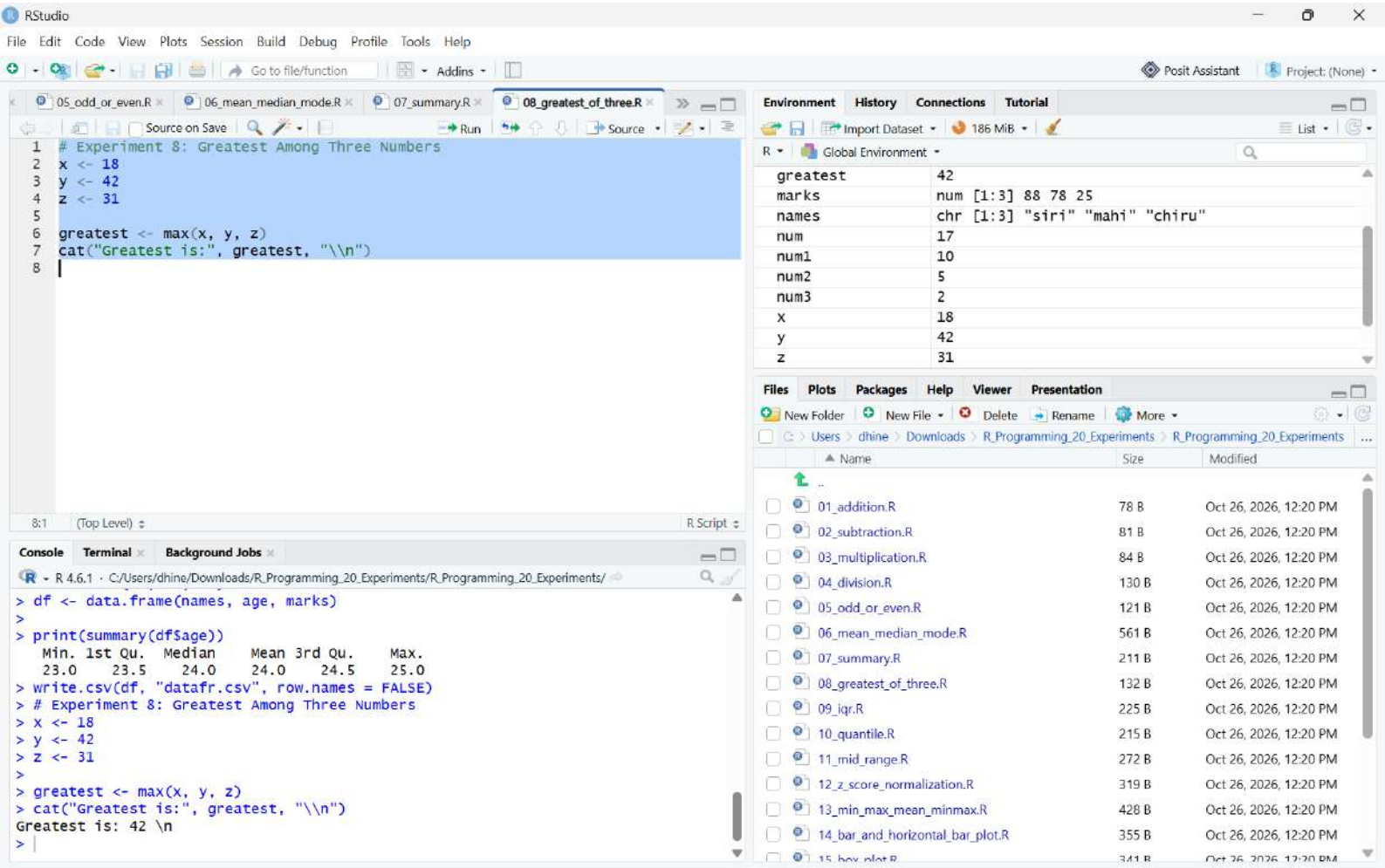

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10_quantile.R
11_mid_range.R
12_z_score_normalization.R
13_min_max_mean_minmax.R
Source on Save
Run
Source
Experiment 13: Minimum, Maximum, Mean, and Min-Max Normalization
age <- c(21, 25, 30, 35, 40)
mean_age <- mean(age)
minimum <- min(age)
maximum <- max(age)
cat("Mean:", mean_age, "\\n")
cat("Minimum:", minimum, "\\n")
cat("Maximum:", maximum, "\\n")
if (maximum == minimum) {
min_max <- rep(0, length(age))
} else {
min_max <- (age - minimum) / (maximum - minimum)
}
cat("Min-max normalized values:\\n")
print(min_max)
19:1
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> maximum <- max(age)
>
> cat("Mean:", mean_age, "\\n")
Mean: 30.2 \n> cat("Minimum:", minimum, "\\n")
Minimum: 21 \n> cat("Maximum:", maximum, "\\n")
Maximum: 40 \n>
> if (maximum == minimum) {
+ min_max <- rep(0, length(age))
+ } else {
+ min_max <- (age - minimum) / (maximum - minimum)
+ }
> cat("Min-max normalized values:\\n")
Min-max normalized values:\n> print(min_max)
[1] 0.0000000 0.2105263 0.4736842 0.7368421 1.0000000
>
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List
R
Global Environment
minimum 21
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 5
num3 2
x 18
y 42
z 31
z_score num [1:5] -1.2112 -0.6846 -0.0263 0.6319 1.2901
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Name
Size
Modified
01_addition.R 78 B Oct 26, 2026, 12:20 PM
02_subtraction.R 81 B Oct 26, 2026, 12:20 PM
03_multiplication.R 84 B Oct 26, 2026, 12:20 PM
04_division.R 130 B Oct 26, 2026, 12:20 PM
05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 561 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_iqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 428 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 355 B Oct 26, 2026, 12:20 PM

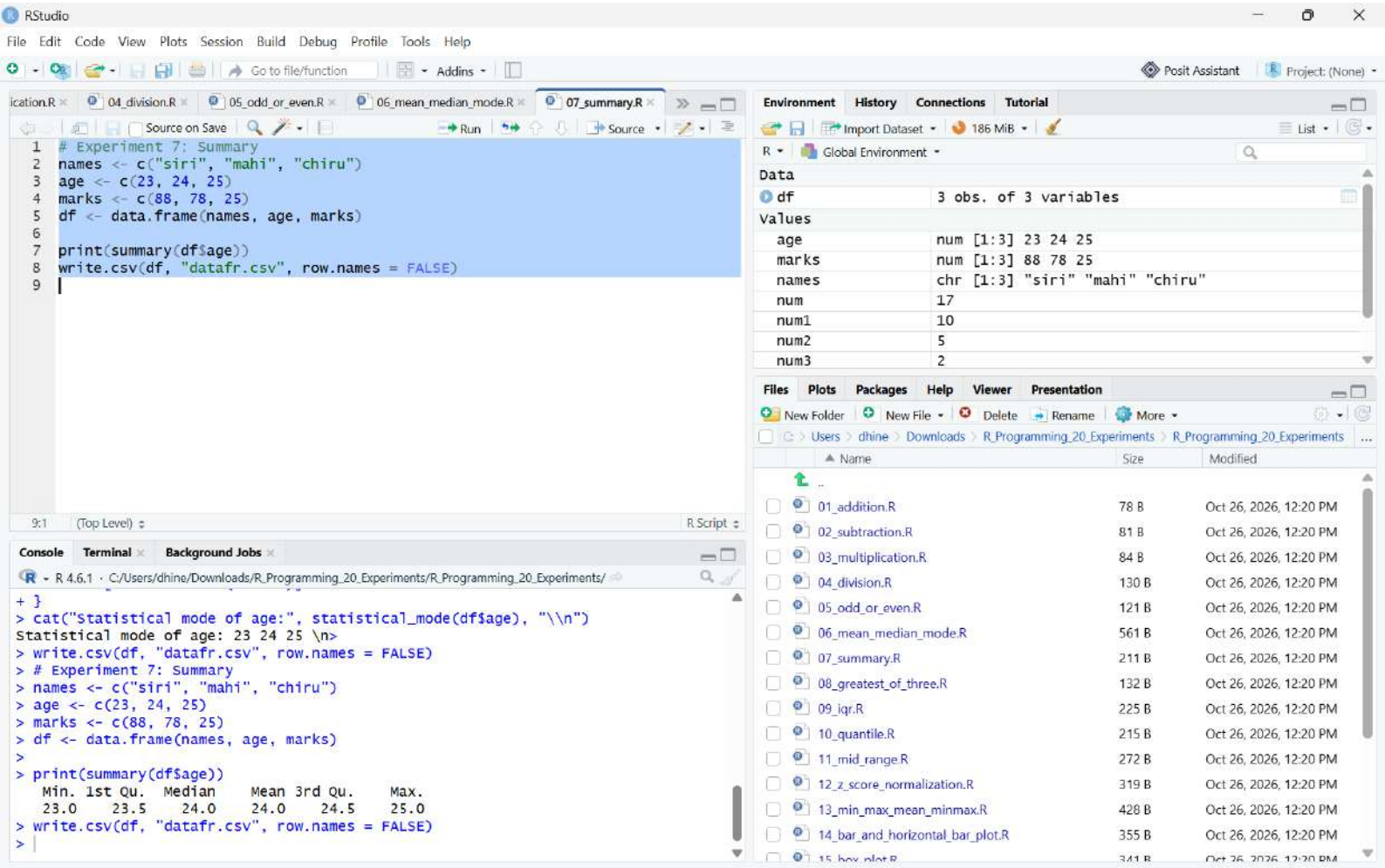

RStudio
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Project: (None)
atest_of_three.R
09_iqr.R
10_quantile.R
11_mid_range.R
12_z_score_normalization.R
Source on Save
Run
Source
1 # Experiment 12: Z-score Normalization
2 # Example Age values are used instead of an external Excel file.
3 age <- c(21, 25, 30, 35, 40)
4 age_mean <- mean(age)
5 age_sd <- sd(age)
6
7 if (age_sd == 0) {
8 stop("Z-score cannot be calculated when standard deviation is zero.")
9 }
10 z_score <- (age - age_mean) / age_sd
11 print(z_score)
12
12:1 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/
> cat("Mid-range:", mid_range, "\\n")
Mid-range: 24 \n> write.csv(df, "datafr.csv", row.names = FALSE)
> # Experiment 12: Z-score Normalization
> # Example Age values are used instead of an external Excel file.
> age <- c(21, 25, 30, 35, 40)
> age_mean <- mean(age)
> age_sd <- sd(age)
>
> if (age_sd == 0) {
+ stop("Z-score cannot be calculated when standard deviation is zero.")
+ }
> z_score <- (age - age_mean) / age_sd
> print(z_score)
[1] -1.21115554 -0.68456618 -0.02632947 0.63190724 1.29014395
>
Environment
History
Connections
Tutorial
Import Dataset
185 MiB
List
R
Global Environment
mid_range 24
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 5
num3 2
x 18
y 42
z 31
z_score num [1:5] -1.2112 -0.6846 -0.0263 0.6319 1.2901
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Users > dhine > Downloads > R_Programming_20_Experiments > R_Programming_20_Experiments
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02_subtraction.R 81 B Oct 26, 2026, 12:20 PM
03_multiplication.R 84 B Oct 26, 2026, 12:20 PM
04_division.R 130 B Oct 26, 2026, 12:20 PM
05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 561 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_iqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 428 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 355 B Oct 26, 2026, 12:20 PM

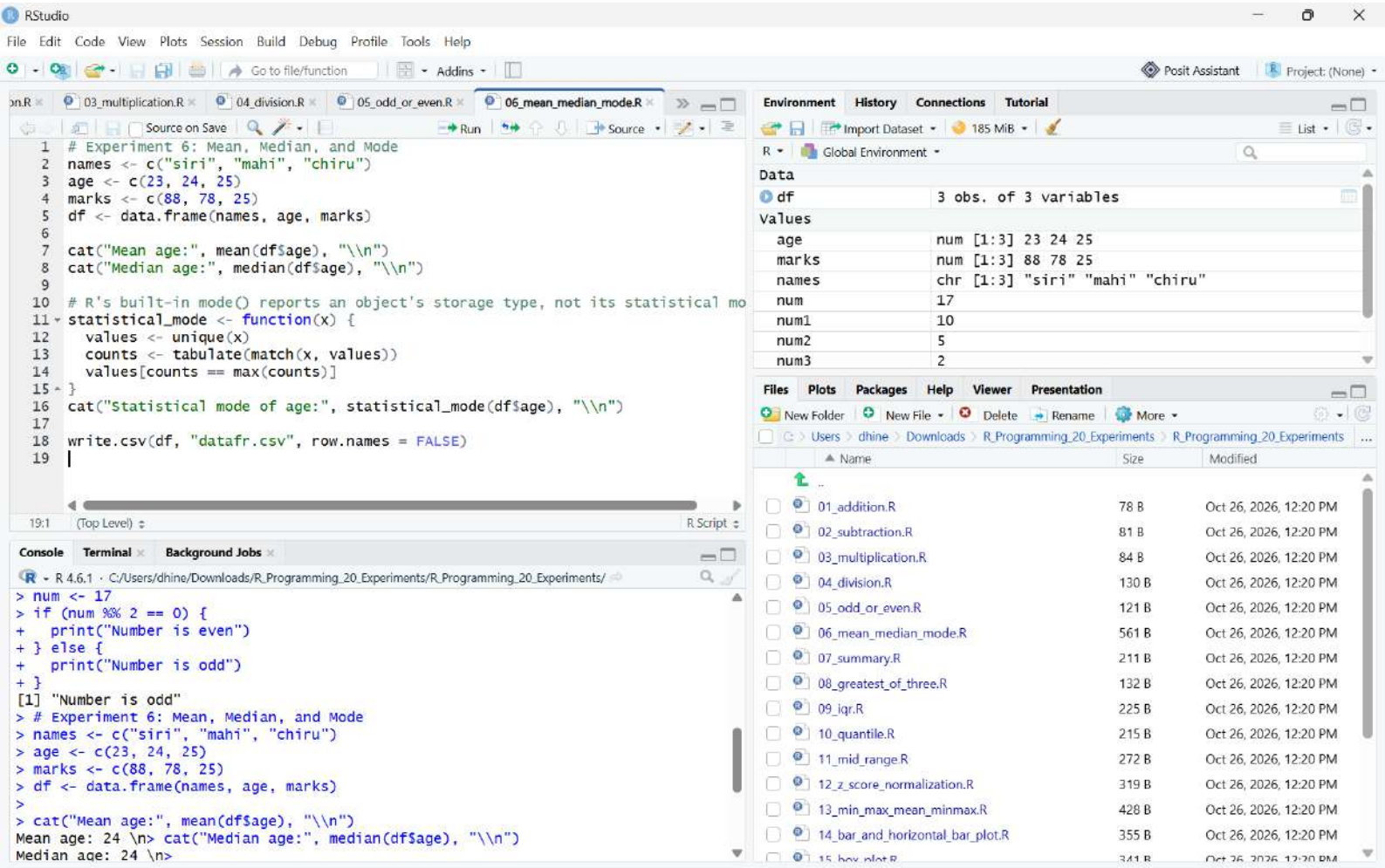

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
07_summary.R
08_greatest_of_three.R
09_iqr.R
10_quantile.R
11_mid_range.R
Experiment 11: Mid-range
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
mid_range <- (min(df$age) + max(df$age)) / 2
cat("Mid-range:", mid_range, "\\n")
write.csv(df, "datafr.csv", row.names = FALSE)
Environment History Connections Tutorial
marks num [1:3] 88 78 25
mid_range 24
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 5
num3 2
x 18
y 42
z 31
Files Plots Packages Help Viewer Presentation
Name Size Modified
01_addition.R 78 B Oct 26, 2026, 12:20 PM
02_subtraction.R 81 B Oct 26, 2026, 12:20 PM
03_multiplication.R 84 B Oct 26, 2026, 12:20 PM
04_division.R 130 B Oct 26, 2026, 12:20 PM
05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 561 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_iqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 428 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 355 B Oct 26, 2026, 12:20 PM
Console Terminal Background Jobs
R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/
> print(quantile(df$age))
0% 25% 50% 75% 100%
23.0 23.5 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)
> # Experiment 11: Mid-range
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> mid_range <- (min(df$age) + max(df$age)) / 2
> cat("Mid-range:", mid_range, "\\n")
Mid-range: 24 \n> write.csv(df, "datafr.csv", row.names = FALSE)

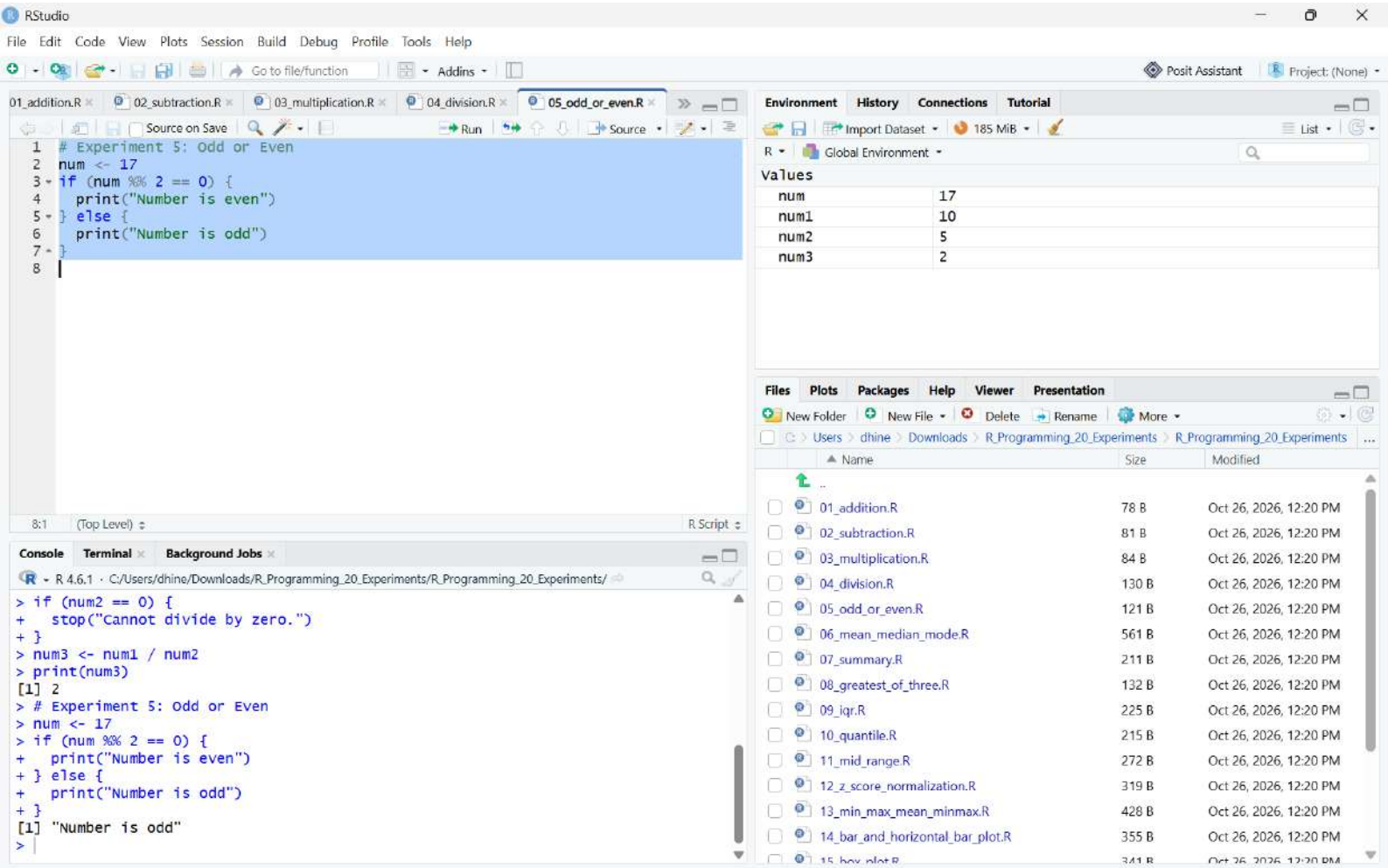

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Posit Assistant
Project: (None)
mean_median_mode.R
07_summary.R
08_greatest_of_three.R
09_iqr.R
10_quantile.R
Source on Save
Run
Source
1 # Experiment 10: Quantiles
2 names <- c("siri", "mahi", "chiru")
3 age <- c(23, 24, 25)
4 marks <- c(88, 78, 25)
5 df <- data.frame(names, age, marks)
6
7 print(quantile(df$age))
8 write.csv(df, "datafr.csv", row.names = FALSE)
9
9:1 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/
> print(quantile(df$age))
0% 25% 50% 75% 100%
23.0 23.5 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)
> # Experiment 10: Quantiles
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
>
> print(quantile(df$age))
0% 25% 50% 75% 100%
23.0 23.5 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)
>
Environment
History
Connections
Tutorial
Import Dataset
185 MiB
List
R
Global Environment
greatest 42
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 5
num3 2
x 18
y 42
z 31
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05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 561 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_iqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 428 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 355 B Oct 26, 2026, 12:20 PM

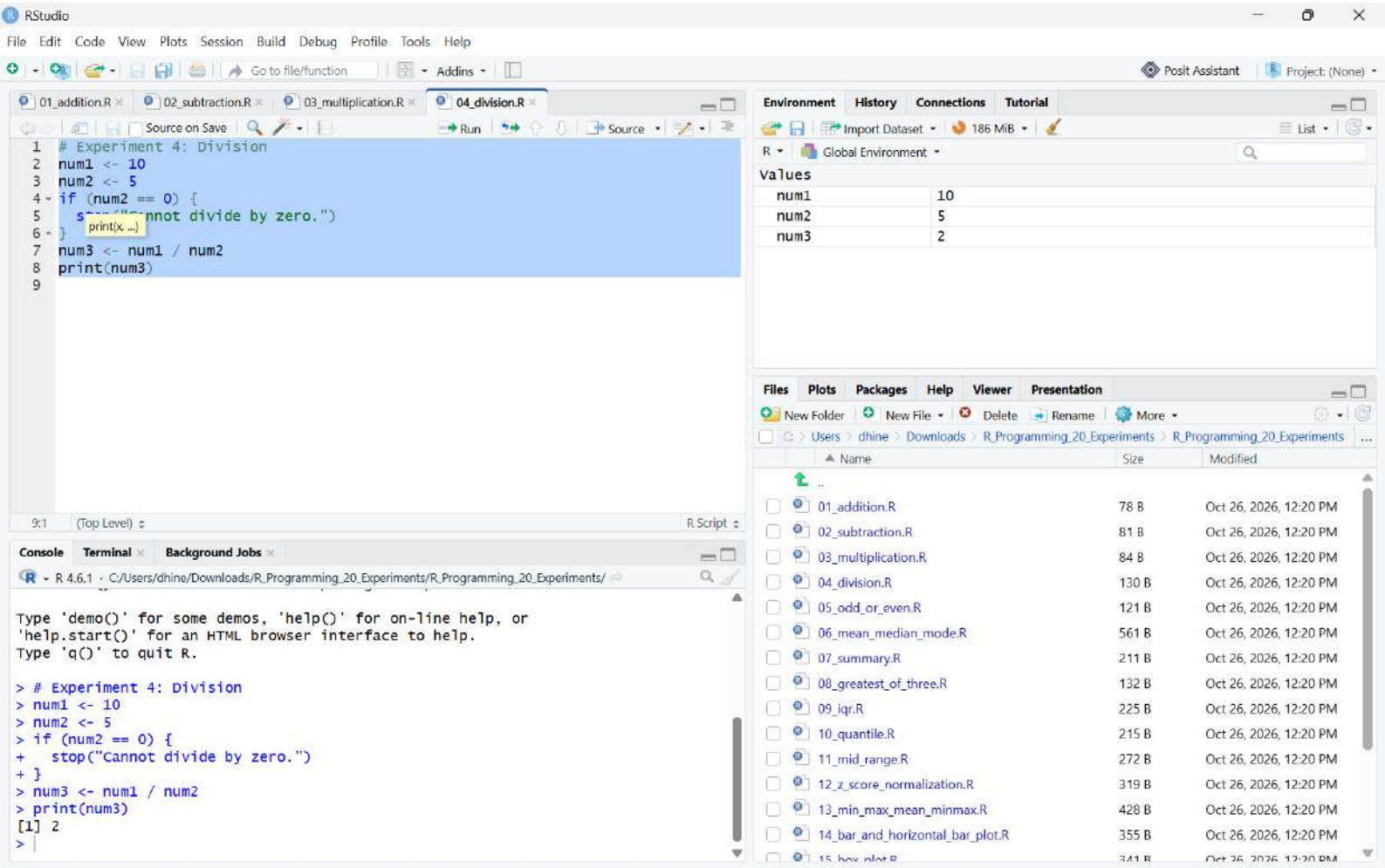

RStudio
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Project: (None)
06_mean_median_mode.R
07_summary.R
08_greatest_of_three.R
09_iqr.R
Source on Save
Run
Source
1 # Experiment 9: Interquartile Range (IQR)
2 names <- c("siri", "mahi", "chiru")
3 age <- c(23, 24, 25)
4 marks <- c(88, 78, 25)
5 df <- data.frame(names, age, marks)
6
7 print(IQR(df$age))
8 write.csv(df, "datafr.csv", row.names = FALSE)
9
9:1 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/
> z <- 31
>
> greatest <- max(x, y, z)
> cat("Greatest is:", greatest, "\\n")
Greatest is: 42 \n
> # Experiment 9: Interquartile Range (IQR)
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
>
> print(IQR(df$age))
[1] 1
> write.csv(df, "datafr.csv", row.names = FALSE)
>
Environment
History
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Tutorial
Import Dataset
185 MiB
List
R
Global Environment
greatest 42
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 5
num3 2
x 18
y 42
z 31
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06_mean_median_mode.R 561 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_iqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 428 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 355 B Oct 26, 2026, 12:20 PM

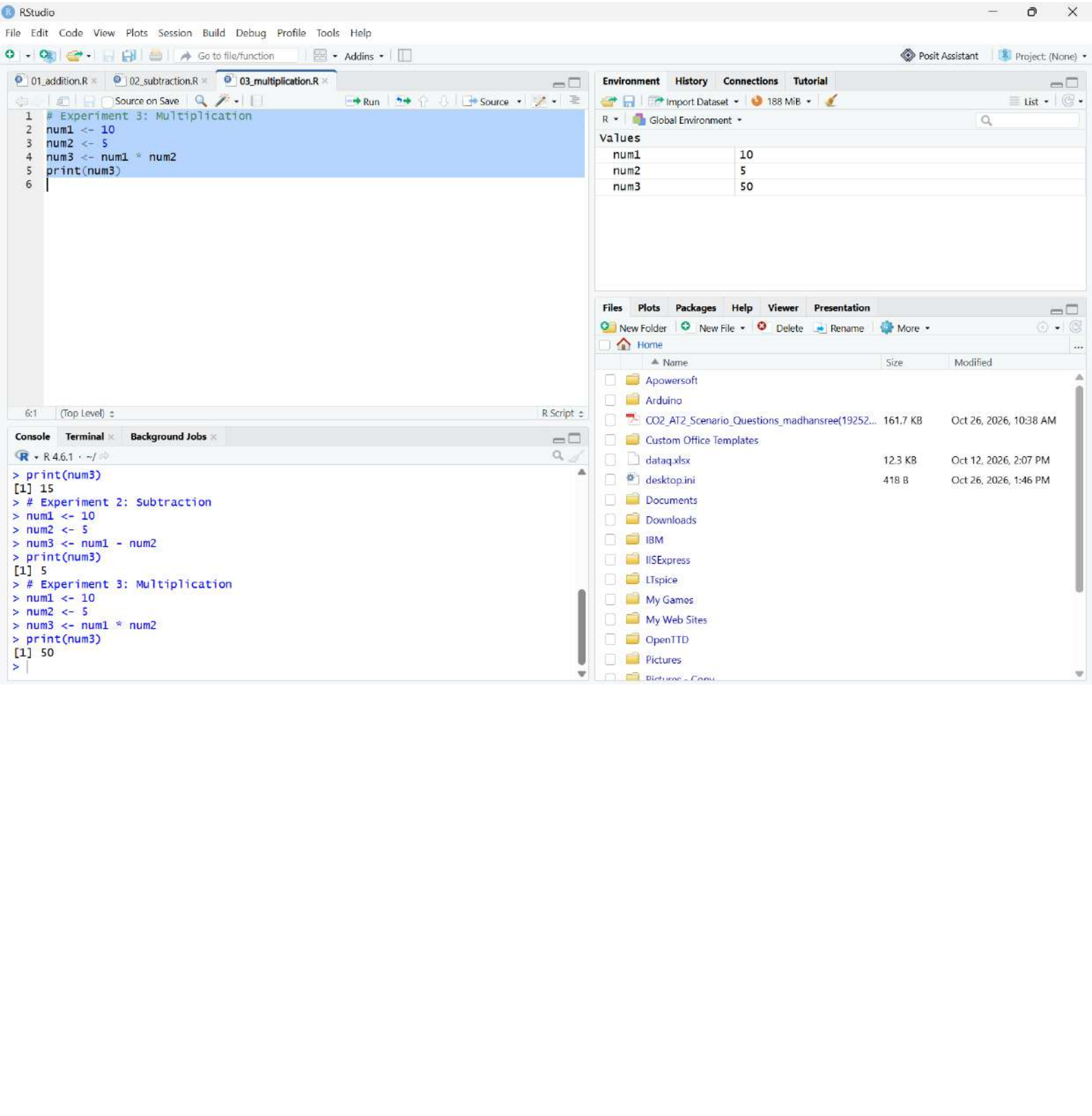

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Project: (None)
05_odd_or_even.R
06_mean_median_mode.R
07_summary.R
08_greatest_of_three.R
Source on Save
Run
Source
1 # Experiment 8: Greatest Among Three Numbers
2 x <- 18
3 y <- 42
4 z <- 31
5
6 greatest <- max(x, y, z)
7 cat("Greatest is:", greatest, "\\n")
8
8:1 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/
> df <- data.frame(names, age, marks)
>
> print(summary(df$age))
Min. 1st Qu. Median Mean 3rd Qu. Max.
23.0 23.5 24.0 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)
> # Experiment 8: Greatest Among Three Numbers
> x <- 18
> y <- 42
> z <- 31
>
> greatest <- max(x, y, z)
> cat("Greatest is:", greatest, "\\n")
Greatest is: 42 \n
>
Environment
History
Connections
Tutorial
Import Dataset
186 MiB
List
R
Global Environment
greatest 42
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 5
num3 2
x 18
y 42
z 31
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Name
Size
Modified
01_addition.R 78 B Oct 26, 2026, 12:20 PM
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03_multiplication.R 84 B Oct 26, 2026, 12:20 PM
04_division.R 130 B Oct 26, 2026, 12:20 PM
05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 561 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_iqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 428 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 355 B Oct 26, 2026, 12:20 PM

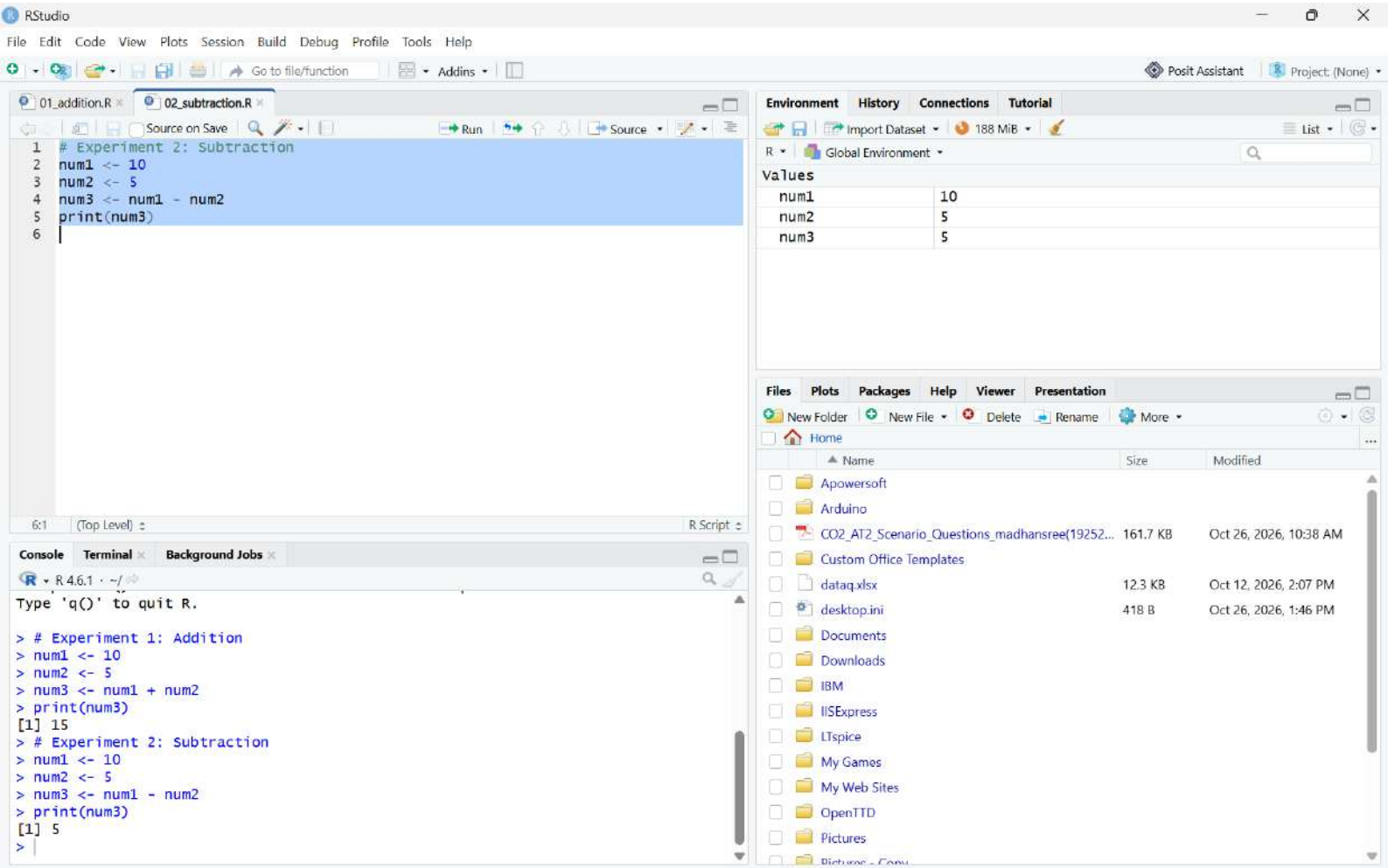

RStudio
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ication.R
04_division.R
05_odd_or_even.R
06_mean_median_mode.R
07_summary.R
Source on Save
Run
Source
1 # Experiment 7: Summary
2 names <- c("siri", "mahi", "chiru")
3 age <- c(23, 24, 25)
4 marks <- c(88, 78, 25)
5 df <- data.frame(names, age, marks)
6
7 print(summary(df$age))
8 write.csv(df, "datafr.csv", row.names = FALSE)
9
9:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/
+ }
> cat("Statistical mode of age:", statistical_mode(df$age), "\\n")
Statistical mode of age: 23 24 25 \n>
> write.csv(df, "datafr.csv", row.names = FALSE)
> # Experiment 7: Summary
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
>
> print(summary(df$age))
Min. 1st Qu. Median Mean 3rd Qu. Max.
23.0 23.5 24.0 24.0 24.5 25.0
> write.csv(df, "datafr.csv", row.names = FALSE)
>
Environment
History
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Import Dataset
186 MiB
List
R
Global Environment
Data
df 3 obs. of 3 variables
Values
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 17
num1 10
num2 5
num3 2
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01_addition.R 78 B Oct 26, 2026, 12:20 PM
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03_multiplication.R 84 B Oct 26, 2026, 12:20 PM
04_division.R 130 B Oct 26, 2026, 12:20 PM
05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 561 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_iqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 428 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 355 B Oct 26, 2026, 12:20 PM
15_box_plot.R 341 B Oct 26, 2026, 12:20 PM

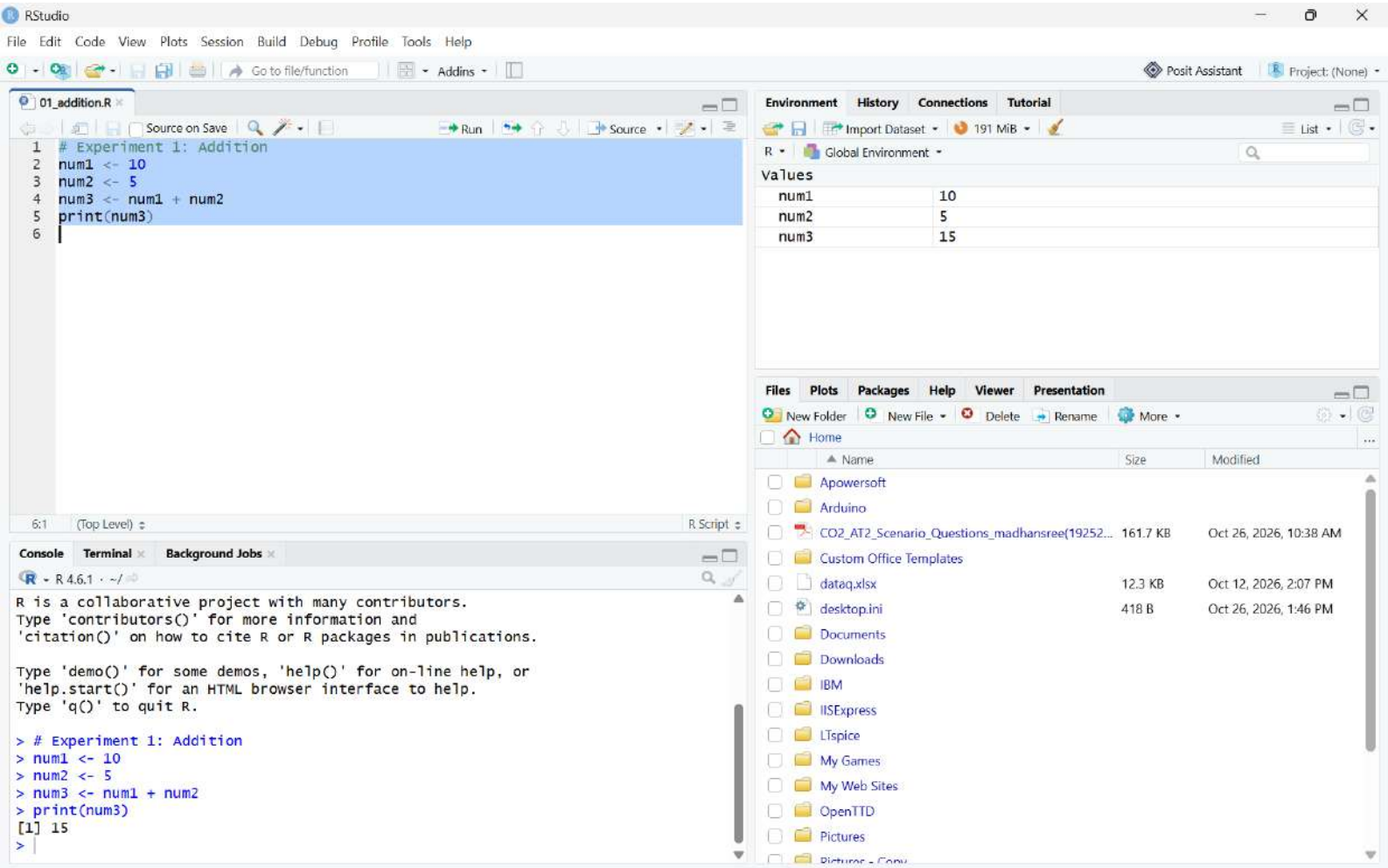

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03_multiplication.R
04_division.R
05_odd_or_even.R
06_mean_median_mode.R
Source on Save
Run
Source
Experiment 6: Mean, Median, and Mode
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
cat("Mean age:", mean(df$age), "\\n")
cat("Median age:", median(df$age), "\\n")
R's built-in mode() reports an object's storage type, not its statistical mo
statistical_mode <- function(x) {
values <- unique(x)
counts <- tabulate(match(x, values))
values[counts == max(counts)]
}
cat("Statistical mode of age:", statistical_mode(df$age), "\\n")
write.csv(df, "datafr.csv", row.names = FALSE)
19:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/
> num <- 17
> if (num %% 2 == 0) {
+ print("Number is even")
+ } else {
+ print("Number is odd")
+ }
[1] "Number is odd"
> # Experiment 6: Mean, Median, and Mode
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
>
> cat("Mean age:", mean(df$age), "\\n")
Mean age: 24 \n> cat("Median age:", median(df$age), "\\n")
Median age: 24 \n>
Environment
History
Connections
Tutorial
Import Dataset
185 MiB
List
R
Global Environment
Data
df
3 obs. of 3 variables
Values
age
num [1:3] 23 24 25
marks
num [1:3] 88 78 25
names
chr [1:3] "siri" "mahi" "chiru"
num
17
num1
10
num2
5
num3
2
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R_Programming_20_Experiments
Name
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03_multiplication.R 84 B Oct 26, 2026, 12:20 PM
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05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 561 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_iqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 428 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 355 B Oct 26, 2026, 12:20 PM


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01_addition.R
02_subtraction.R
03_multiplication.R
04_division.R
05_odd_or_even.R
Source on Save
Run
Source
1 # Experiment 5: Odd or Even
2 num <- 17
3 if (num %% 2 == 0) {
4 print("Number is even")
5 } else {
6 print("Number is odd")
7 }
8
8:1 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/
> if (num2 == 0) {
+ stop("Cannot divide by zero.")
+ }
> num3 <- num1 / num2
> print(num3)
[1] 2
> # Experiment 5: Odd or Even
> num <- 17
> if (num %% 2 == 0) {
+ print("Number is even")
+ } else {
+ print("Number is odd")
+ }
[1] "Number is odd"
>
Environment
History
Connections
Tutorial
Import Dataset
185 MiB
List
R
Global Environment
Values
num 17
num1 10
num2 5
num3 2
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Size
Modified
01_addition.R 78 B Oct 26, 2026, 12:20 PM
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03_multiplication.R 84 B Oct 26, 2026, 12:20 PM
04_division.R 130 B Oct 26, 2026, 12:20 PM
05_odd_or_even.R 121 B Oct 26, 2026, 12:20 PM
06_mean_median_mode.R 561 B Oct 26, 2026, 12:20 PM
07_summary.R 211 B Oct 26, 2026, 12:20 PM
08_greatest_of_three.R 132 B Oct 26, 2026, 12:20 PM
09_iqr.R 225 B Oct 26, 2026, 12:20 PM
10_quantile.R 215 B Oct 26, 2026, 12:20 PM
11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 428 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 355 B Oct 26, 2026, 12:20 PM


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04_division.R
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Source
Experiment 4: Division
num1 <- 10
num2 <- 5
if (num2 == 0) {
stop("Cannot divide by zero.")
}
num3 <- num1 / num2
print(num3)
print(x, ...)
9:1
(Top Level)
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04_division.R 130 B Oct 26, 2026, 12:20 PM
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06_mean_median_mode.R 561 B Oct 26, 2026, 12:20 PM
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11_mid_range.R 272 B Oct 26, 2026, 12:20 PM
12_z_score_normalization.R 319 B Oct 26, 2026, 12:20 PM
13_min_max_mean_minmax.R 428 B Oct 26, 2026, 12:20 PM
14_bar_and_horizontal_bar_plot.R 355 B Oct 26, 2026, 12:20 PM
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R 4.6.1 · C:/Users/dhine/Downloads/R_Programming_20_Experiments/R_Programming_20_Experiments/
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
> # Experiment 4: Division
> num1 <- 10
> num2 <- 5
> if (num2 == 0) {
+ stop("Cannot divide by zero.")
+ }
> num3 <- num1 / num2
> print(num3)
[1] 2
>


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Source
1 # Experiment 3: Multiplication
2 num1 <- 10
3 num2 <- 5
4 num3 <- num1 * num2
5 print(num3)
6
6:1 (Top Level)
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Background Jobs
R 4.6.1 · ~/
> print(num3)
[1] 15
> # Experiment 2: Subtraction
> num1 <- 10
> num2 <- 5
> num3 <- num1 - num2
> print(num3)
[1] 5
> # Experiment 3: Multiplication
> num1 <- 10
> num2 <- 5
> num3 <- num1 * num2
> print(num3)
[1] 50
>
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dataq.xlsx 12.3 KB Oct 12, 2026, 2:07 PM
desktop.ini 418 B Oct 26, 2026, 1:46 PM
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1 # Experiment 2: Subtraction
2 num1 <- 10
3 num2 <- 5
4 num3 <- num1 - num2
5 print(num3)
6
6:1
(Top Level)
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R 4.6.1 · ~/
Type 'q()' to quit R.
> # Experiment 1: Addition
> num1 <- 10
> num2 <- 5
> num3 <- num1 + num2
> print(num3)
[1] 15
> # Experiment 2: Subtraction
> num1 <- 10
> num2 <- 5
> num3 <- num1 - num2
> print(num3)
[1] 5
>


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Source
1 # Experiment 1: Addition
2 num1 <- 10
3 num2 <- 5
4 num3 <- num1 + num2
5 print(num3)
6
6:1 (Top Level)
R Script
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R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
> # Experiment 1: Addition
> num1 <- 10
> num2 <- 5
> num3 <- num1 + num2
> print(num3)
[1] 15
>
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